

大偶数表为一个素数及一个不超过二个素数的乘积之和

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摘要

本文的目的在于用筛法证明了：每一充分大的偶数是一个素数及一个不超过两个素数乘积之和。

关于孪生素数问题亦得到类似的结果。

1 引言

把命题“每一个充分大的偶数都能表示为一个素数及一个不超过 a 个素数的乘积之和”简记为 $(1, a)$ 。

不少数学工作者改进了筛法及素数分布的某些结果，并用以改善 $(1, a)$ 。现在我们将 $(1, a)$ 发展历史简述如下：

- $(1, c)$ ——Renyi^[1],
- $(1, 5)$ ——潘承洞^[2]、Барбан^[3],
- $(1, 4)$ ——王元^[4]、潘承洞^[5]、Барбан^[6],
- $(1, 3)$ ——Бухштаб^[7]、Виноградов^[8]、Bombieri^[9],

在文献 [10] 中我们给出了 $(1, 2)$ 的证明提要。

命 $P_x(1, 2)$ 为适合下列条件的素数 p 的个数：

$$x - p = p_1 \quad \text{或} \quad x - p = p_2 p_3,$$

其中 p_1, p_2, p_3 都是素数。

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用 x 表一充分大的偶数。命

$$C_x = \prod_{\substack{p|x \\ p>2}} \frac{p-1}{p-2} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right).$$

对于任意给定的偶数 h 及充分大的 x , 用 $x_h(1, 2)$ 表示满足下面条件的素数 p 的个数:

$$p \leq x, \quad p+h = p_1 \quad \text{或} \quad p+h = p_2 p_3,$$

其中 p_1, p_2, p_3 都是素数。

本文目的在于证明并改进作者在文献 [10] 内所提及的全部结果, 现在详述如下。

定理 1. $(1, 2)$ 及 $P_x(1, 2) \geq \frac{0.67xC_x}{(\log x)^2}$ 。

定理 2. 对于任意偶数 h , 都存在无限多个素数 p , 使得 $p+h$ 的素因子的个数不超过 2 个及

$$x_h(1, 2) \geq \frac{0.67xC_x}{(\log x)^2}.$$

在证明定理 1 时, 主要用到了本文中的引理 8 和引理 9。在证明引理 8 时, 我们使用较为简单的数字计算方法; 而证明引理 9 时, 我们使用了 Bombieri 定理 [9] 及 Richert [11] 中的一个结果。

2 几个引理

引理 1. 假设 $y \geq 0$, 而 $[\log x]$ 表示 $\log x$ 的整数部分, $x > 1$,

$$\Phi(y) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \frac{y^\omega d\omega}{\omega \left(1 + \frac{\omega}{(\log x)^{1.1}}\right)^{[\log x]+1}}.$$

显见, 当 $0 \leq y \leq 1$ 时, 有 $\Phi(y) = 0$ 。对于所有 $y \geq 0$, 则 $\Phi(y)$ 是一个非减函数。当 $\log x \geq 10^4$ 及 $y \geq e^{2(\log x)^{-0.1}}$ 时, 则有

$$1 - x^{-0.1} \leq \Phi(y) \leq 1.$$

证. 我们先来证明

$$\frac{\partial^r}{\partial \omega^r} \left(\frac{y^\omega}{\omega} \right) = \left(\frac{y^\omega}{\omega} \right) \left\{ (\log y)^r + \sum_{i=1}^r \frac{(-1)^i r \cdots (r-i+1) (\log y)^{r-i}}{\omega^i} \right\} \quad (1)$$

成立。显见, (1) 式当 $r = 1$ 和 $r = 2$ 时都成立。现假定 (1) 式对于 $r = 2, \dots, S$ 时都成立, 而证明对于 $S + 1$ 也成立。由于

$$\begin{aligned}
 \frac{\partial^{S+1}}{\partial \omega^{S+1}} \left(\frac{y^\omega}{\omega} \right) &= \frac{\partial}{\partial \omega} \left\{ y^\omega \left(\frac{(\log y)^S}{\omega} + \sum_{i=1}^S \frac{(-1)^i S \cdots (S-i+1)(\log y)^{S-i}}{\omega^{i+1}} \right) \right\} \\
 &= y^\omega \left\{ \frac{(\log y)^{S+1}}{\omega} + \sum_{i=1}^S \frac{(-1)^i S \cdots (S-i+1)(\log y)^{S+1-i}}{\omega^{i+1}} - \frac{(\log y)^S}{\omega^2} \right. \\
 &\quad \left. + \sum_{i=1}^S \frac{(-1)^{i+1} S \cdots (S-i+1)(i+1)(\log y)^{S-i}}{\omega^{i+2}} \right\} \\
 &= \left(\frac{y^\omega}{\omega} \right) \left\{ (\log y)^{S+1} - \frac{(S+1)(\log y)^S}{\omega} + \frac{(-1)^{S+1}(S+1)!}{\omega^{S+1}} \right. \\
 &\quad \left. + \sum_{i=2}^S \left(\frac{(-1)^i S \cdots (S-i+1)(\log y)^{S+1-i}}{\omega^i} + \frac{(-1)^i S \cdots (S+2-i)i(\log y)^{S+1-i}}{\omega^i} \right) \right\} \\
 &= \left(\frac{y^\omega}{\omega} \right) \left\{ (\log y)^{S+1} + \sum_{i=1}^{S+1} \frac{(-1)^i (S+1) \cdots (S+1-i+1)(\log y)^{S+1-i}}{\omega^i} \right\}.
 \end{aligned}$$

故 (1) 式得证。

又当 $y \geq 1$ 时, 我们有

$$\begin{aligned}
 \Phi(y) &= 1 + \left\{ \frac{(\log x)^{1.1+1.1[\log x]}}{[\log x]!} \right\} \left\{ \frac{\partial^{[\log x]}}{\partial \omega^{[\log x]}} \left(\frac{y^\omega}{\omega} \right) \right\}_{\omega=-(\log x)^{1.1}} \\
 &= 1 - e^{-(\log x)^{1.1}(\log y)} \sum_{\nu=0}^{[\log x]} \frac{\{(\log x)^{1.1}(\log y)\}^\nu}{\nu!} \\
 &= \left\{ \frac{1}{[\log x]!} \right\} \int_0^{(\log x)^{1.1}(\log y)} e^{-\lambda} \lambda^{[\log x]} d\lambda.
 \end{aligned}$$

因为 $0 \leq y \leq 1$ 时, $\Phi(y) = 0$ 。故由上式得到: 当 $y \geq 0$ 时, 则 $\Phi(y)$ 是一个非减函数。

又当 $y \geq e^{2(\log x)^{-0.1}}$ 时, 有

$$\begin{aligned}
 0 < 1 - \Phi(y) &= \left\{ \frac{1}{[\log x]!} \right\} \int_{(\log x)^{1.1}(\log y)}^{\infty} e^{-\lambda} \lambda^{[\log x]} d\lambda \\
 &\leq \left\{ \frac{1}{[\log x]!} \right\} \int_{2[\log x]}^{\infty} e^{-\lambda} \lambda^{[\log x]} d\lambda = \left\{ \frac{\{[\log x]\}^{1+[\log x]}}{[\log x]!} \right\} \\
 &\quad \cdot \int_2^{\infty} e^{-\lambda[\log x]} \lambda^{[\log x]} d\lambda = \left\{ \frac{e^{-[\log x]} \{[\log x]\}^{1+[\log x]}}{[\log x]!} \right\} \\
 &\quad \cdot \int_1^{\infty} e^{-\lambda[\log x]} (1+\lambda)^{[\log x]} d\lambda \leq x^{-0.1}.
 \end{aligned}$$

其中用到 $\log x \geq 10^4$ 及当 $\lambda \geq 1$ 时, 有 $e^{\log(1+\lambda)} \leq e^{\lambda \log 2}$ 。

□

引理 2. 令 $e(\alpha) = e^{2\pi i\alpha}$, $S(\alpha) = \sum_{n=M+1}^{M+N} a_n e(n\alpha)$, $Z = \sum_{n=M+1}^{M+N} |a_n|^2$, 其中 a_n 是任意的实数。我们用 $\sum_{\chi_q}^*$ 来表示和式之中经过且只经过模 q 的所有原特征, 则有

$$\sum_{q \leq X} \frac{q}{\varphi(q)} \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 \leq (X^2 + \pi N) \sum_{n=M+1}^{M+N} |a_n|^2; \quad (2)$$

$$\sum_{D < q \leq Q} \frac{1}{\varphi(q)} \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 \ll \left(Q + \frac{N}{D} \right) \sum_{n=M+1}^{M+N} |a_n|^2. \quad (3)$$

证. 令 F 是一个周期为 1 的复数值可微函数, 则有

$$\left| F\left(\frac{a}{q}\right) \right| = \left| F(\alpha) - \int_{\frac{a}{q}}^{\alpha} dF(\beta) \right| \leq |F(\alpha)| + \int_{\frac{a}{q}}^{\alpha} |F'(\beta)| d\beta,$$

我们用 $I(a, q)$ 来表示以 $\frac{a}{q}$ 为中心, 而长度 $\frac{1}{Q^2}$ 的区间。显见, 当 $1 \leq a < q$, $(a, q) = 1$, $q \leq Q$ 时, 所有的区间 $I(a, q)$ 都没有共同部分, 故得

$$\begin{aligned} \sum_{q \leq Q} \sum_{\substack{a=1 \\ (a, q)=1}}^q \left| F\left(\frac{a}{q}\right) \right| &\leq \sum_{q \leq Q} \sum_{\substack{a=1 \\ (a, q)=1}}^q \left\{ Q^2 \int_{I(a, q)} |F(\alpha)| d\alpha + \frac{1}{2} \int_{I(a, q)} |F'(\beta)| d\beta \right\} \\ &\leq Q^2 \int_0^1 |F(\alpha)| d\alpha + \frac{1}{2} \int_0^1 |F'(\beta)| d\beta. \end{aligned}$$

我们取 $F(\alpha) = \{S(\alpha)\}^2$, 则得

$$\int_0^1 |F(\alpha)| d\alpha = Z$$

及

$$\begin{aligned} \frac{1}{2} \int_0^1 |F'(\beta)| d\beta &= \int_0^1 |S(\alpha)| |S'(\alpha)| d\alpha \\ &\leq \left\{ \left(\int_0^1 |S(\alpha)|^2 d\alpha \right) \left(\int_0^1 |S'(\alpha)|^2 d\alpha \right) \right\}^{\frac{1}{2}} = Z^{\frac{1}{2}} \left(\int_0^1 |S'(\alpha)|^2 d\alpha \right)^{\frac{1}{2}}. \end{aligned}$$

故有

$$\begin{aligned}
\sum_{q \leq Q} \sum_{\substack{a=1 \\ (a,q)=1}}^q \left| S\left(\frac{a}{q}\right) \right|^2 &= \sum_{q \leq Q} \sum_{\substack{a=1 \\ (a,q)=1}}^q \left\{ \left| S\left(\frac{a}{q}\right) \right| e\left(-\frac{a(M + [\frac{N}{2}])}{q}\right) \right\}^2 \\
&= \sum_{q \leq Q} \sum_{\substack{a=1 \\ (a,q)=1}}^q \left| \sum_{n=M+1}^{M+N} a_n e\left(\left\{n - \left(M + \left[\frac{N}{2}\right]\right)\right\} \frac{a}{q}\right) \right|^2 \\
&= \sum_{q \leq Q} \sum_{\substack{a=1 \\ (a,q)=1}}^q \left| \sum_{-\lceil \frac{N}{2} \rceil + 1 \leq n \leq N - \lceil \frac{N}{2} \rceil} a_{n+M+\lceil \frac{N}{2} \rceil} e\left(\frac{na}{q}\right) \right|^2 \\
&\leq ZQ^2 + Z^{\frac{1}{2}} \left\{ \sum_{n=-\lceil \frac{N}{2} \rceil + 1}^{N - \lceil \frac{N}{2} \rceil} ((2\pi n) a_{n+M+\lceil \frac{N}{2} \rceil})^2 \right\}^{\frac{1}{2}} \\
&\leq ZQ^2 + \pi N Z^{\frac{1}{2}} \left(\sum_{n=-\lceil \frac{N}{2} \rceil + 1}^{N - \lceil \frac{N}{2} \rceil} |a_{n+M+\lceil \frac{N}{2} \rceil}|^2 \right)^{\frac{1}{2}} \\
&\leq (Q^2 + \pi N)Z.
\end{aligned} \tag{4}$$

令 χ^* 表示原特征,

$$\tau(\chi_q^*) = \sum_{1 \leq a < q} \chi_q^*(a) e\left(\frac{a}{q}\right), \quad \tau(\overline{\chi_q^*}) \chi_q^*(n) = \sum_{a=1}^q \overline{\chi_q^*}(a) e\left(\frac{na}{q}\right).$$

由于 $|\tau(\overline{\chi_q^*})|^2 = q$, 故得到

$$\begin{aligned}
\left(\frac{1}{\varphi(q)}\right) \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 &\leq \left(\frac{1}{q\varphi(q)}\right) \sum_{\chi_q}^* \left| \tau(\overline{\chi_q}) \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 \\
&= \left(\frac{1}{q\varphi(q)}\right) \sum_{\chi_q}^* \left| \sum_{a=1}^q \overline{\chi_q}(a) \sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right) \right|^2 \\
&\leq \left(\frac{1}{q\varphi(q)}\right) \sum_{\chi_q} \left| \sum_{a=1}^q \overline{\chi_q}(a) \sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right) \right|^2 \\
&\leq \frac{1}{q} \sum_{\substack{a=1 \\ (a,q)=1}}^q \left| \sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right) \right|^2.
\end{aligned}$$

由上式及 (4) 式, 即得到 (2) 式。我们定义 h 是一个正整数, 它使得 $2^h D < Q \leq$

$2^{h+1}D$, 则我们有

$$\begin{aligned}
 \sum_{D < q \leq Q} \frac{1}{\varphi(q)} \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 &\leq \sum_{i=0}^h \left(\sum_{2^i D < q \leq 2^{i+1} D} \frac{1}{\varphi(q)} \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 \right) \\
 &\leq \sum_{i=0}^h \left(\frac{1}{2^i D} \right) \left(\sum_{2^i D < q \leq 2^{i+1} D} \frac{q}{\varphi(q)} \sum_{\chi_q}^* \left| \sum_{n=M+1}^{M+N} a_n \chi_q(n) \right|^2 \right) \\
 &\leq \sum_{i=0}^h \left(2^{i+2} D + \frac{\pi N}{2^i D} \right) \sum_{n=M+1}^{M+N} |a_n|^2 \\
 &\ll \left(Q + \frac{N}{D} \right) \sum_{n=M+1}^{M+N} |a_n|^2.
 \end{aligned}$$

故引理 2 得证。 $\square$

引理 3. 当 $S = \sigma + it$ 和 $\sigma \geq \frac{1}{2}$ 时, 则有

$$\sum_{q \leq Q} \sum_{\chi_q}^* |L(S, \chi_q)|^4 \ll Q^2 |S|^2 (\log Q)^4.$$

证. 我们有

$$\begin{aligned}
 L(S, \chi) &= \sum_{n=1}^{\infty} \frac{\chi(n)}{n^S} = \sum_{n=1}^N \frac{\chi(n)}{n^S} + \sum_{n=N+1}^{\infty} \frac{\sum_{i \leq n} \chi(i) - \sum_{i \leq n-1} \chi(i)}{n^S} \\
 &= \sum_{n=1}^N \frac{\chi(n)}{n^S} + \sum_{n=N+1}^{\infty} \left(\sum_{i \leq n} \chi(i) \right) \left(\frac{1}{n^S} - \frac{1}{(n+1)^S} \right) - \frac{\sum_{i \leq N} \chi(i)}{(N+1)^S} \\
 &= \sum_{n=1}^N \frac{\chi(n)}{n^S} + O \left(\frac{|S| q^{\frac{1}{2}} \log q}{N^\sigma} \right).
 \end{aligned}$$

故由引理 2 及 $\sigma \geq \frac{1}{2}$, 我们有

$$\begin{aligned}
 \sum_{q \leq Q} \sum_{\chi_q}^* |L(S, \chi_q)|^4 &\ll \sum_{q \leq Q} \sum_{\chi_q}^* \left(\left| \sum_{n=1}^{[Q|S|]} \frac{\chi_q(n)}{n^S} \right|^4 + Q^{-2} |S|^2 q^2 (\log q)^4 \right) \\
 &\ll |S|^2 Q^2 (\log Q)^4 + (Q^2 + Q^2 |S|^2) \sum_{n=1}^{[Q|S|]^2} \frac{d^2(n)}{n} \\
 &\ll Q^2 |S|^2 (\log Q)^4.
 \end{aligned}$$

故本引理得证。 $\square$

引理 4. 当 k 是无平方因子的奇数, 而 $m \neq 1$ 时, 则我们有

$$\left| \sum_{\chi_k}^* \chi_k(m) \right| \leq |(m-1, k)|.$$

证. 令 $k = p_1 \cdots p_l$, 而 $p_1 < \cdots < p_l$. 令 g_i 是 $\text{mod } p_i$ 的原根, 则有 $m \equiv g_i^{\xi_i} \pmod{p_i}$, $0 \leq \xi_i \leq p_i - 2$, $j = 1, \dots, l$, 则关于模 k 的所有原特征可表示为

$$\chi_k^*(m) = e^{2\pi i \left(\frac{\nu_1 \xi_1}{p_1-1} + \cdots + \frac{\nu_l \xi_l}{p_l-1} \right)},$$

其中 $1 \leq \nu_i \leq p_i - 2$, 而 $i = 1, \dots, l$.

令 $Z(m, k) = \left| \sum_{\chi_k}^* \chi_k(m) \right|$, 则有

$$Z(m, k) = \prod_{j=1}^l Z(m, p_j) = \prod_{j=1}^l \left| \sum_{\nu_j=1}^{p_j-2} e^{2\pi i \frac{\nu_j \xi_j}{p_j-1}} \right| = \prod_{\substack{j=1 \\ \xi_j=0}}^l (p_j - 2) < \prod_{p_j | (m-1)} p_j = |(m-1, k)|.$$

故本引理得证. $\square$

设 x 是偶数, 令 $\lambda_1 = 1$; 当 $d > x^{\frac{1}{4}-\frac{\epsilon}{2}}$ 时, 令 $\lambda_d = 0$; 而当 $1 < d \leq x^{\frac{1}{4}-\frac{\epsilon}{2}}$ 时, 令

$$\lambda_d = \frac{\mu(d)}{f(d)g(d)} \left\{ \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k, xd)=1}} \frac{\mu^2(k)}{f(k)} \right\} \left\{ \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k, x)=1}} \frac{\mu^2(k)}{f(k)} \right\}^{-1}.$$

其中 $g(k) = \frac{1}{\varphi(k)}$, $f(k) = \varphi(k) \prod_{p|k} \frac{p-2}{p-1}$. 又当 d 为奇数, $\mu(d) \neq 0$ 时, 有

$$\begin{aligned} \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k, x)=1}} \frac{\mu^2(k)}{f(k)} &= \sum_{t|d} \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k, x)=1, (k, d)=t}} \frac{\mu^2(k)}{f(k)} = \sum_{t|d} \left\{ \frac{1}{\prod_{p|t} (p-2)} \right\} \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/t \\ (k, xd)=1}} \frac{\mu^2(k)}{f(k)} \\ &\geq \left\{ \prod_{p|d} \left(1 + \frac{1}{p-2} \right) \right\} \left\{ \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k, xd)=1}} \frac{\mu^2(k)}{f(k)} \right\}. \end{aligned}$$

故对于所有正整数 d , 都有 $|\lambda_d| \leq 1$. 设 x 是偶数, $\log x > 10^4$, 又令

$$\begin{aligned} Q &= \prod_{2 \leq p \leq x^{\frac{1}{4}}} p, \\ \Omega &= \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ p_3 \leq \frac{x}{p_1 p_2} \\ (x - p_1 p_2 p_3, Q) = 1}} 1, \\ M &= \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \left(\sum_{\substack{n \leq \frac{x}{p_1 p_2} \\ (x - p_1 p_2 n, Q) = 1}} \Lambda(n) \right), \end{aligned}$$

则有 $\Omega \leq \frac{M}{1-\epsilon} + N$, 其中

$$\begin{aligned} N &\ll \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{x}{p_1 p_2}\right)^{1-\epsilon} \ll x^{1-\epsilon} \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S^{1-\epsilon}} \int_{x^{\frac{1}{3}}}^{(\frac{x}{S})^{\frac{1}{2}}} \frac{dt}{t^{1-\epsilon}} \\ &\ll x^{1-\frac{\epsilon}{2}} \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S^{1-\frac{\epsilon}{2}}} \ll x^{1-\frac{\epsilon}{3}}. \end{aligned}$$

由引理 1, 我们有

$$\begin{aligned} M &\leq \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{1}{\log \frac{x}{p_1 p_2}}\right) \sum_{\substack{n \leq \frac{x}{p_1 p_2} \\ (x-p_1 p_2 n, Q)=1}} \Lambda(n) \Phi\left(\frac{x}{p_1 p_2 n}\right) + O\left(\frac{x}{(\log x)^{2.01}}\right) \\ &\leq \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{1}{\log \frac{x}{p_1 p_2}}\right) \sum_{n \leq \frac{x}{p_1 p_2}} \Lambda(n) \Phi\left(\frac{x}{p_1 p_2 n}\right) \left(\sum_{\substack{d|(x-p_1 p_2 n, Q) \\ (d, x)=1}} \lambda_d\right)^2 + O\left(\frac{x}{(\log x)^{2.01}}\right) \\ &= \sum_{\substack{(d_1, x)=1 \\ d_1|Q}} \sum_{\substack{(d_2, x)=1 \\ d_2|Q}} \lambda_{d_1} \lambda_{d_2} N_{\frac{d_1 d_2}{(d_1, d_2)}} + O\left(\frac{x}{(\log x)^{2.01}}\right). \end{aligned} \tag{5}$$

其中

$$\begin{aligned}
N_{\frac{d_1 d_2}{(d_1, d_2)}} &= \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \sum_{\substack{n \leq \frac{x}{p_1 p_2} \\ x - p_1 p_2 n \equiv 0 \pmod{\frac{d_1 d_2}{(d_1, d_2)}}}} \Lambda(n) \Phi \left(\frac{x}{p_1 p_2 n} \right) \\
&= \left\{ \frac{1}{\varphi \left(\frac{d_1 d_2}{(d_1, d_2)} \right)} \right\} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2} \\ (p_1 p_2 n, d_1 d_2) = 1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Lambda(n) \Phi \left(\frac{x}{p_1 p_2 n} \right) \\
&\quad + \sum_{\chi_{\frac{d_1 d_2}{(d_1, d_2)}} \neq \chi_0} \overline{\chi_{\frac{d_1 d_2}{(d_1, d_2)}}}(x) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2}}} \left(\frac{\Lambda(n)}{\log \frac{x}{p_1 p_2}} \right) \Phi \left(\frac{x}{p_1 p_2 n} \right) \chi_{\frac{d_1 d_2}{(d_1, d_2)}}(p_1 p_2 n) \\
&= \left\{ \frac{1}{\varphi \left(\frac{d_1 d_2}{(d_1, d_2)} \right)} \right\} \left\{ \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2} \\ (p_1 p_2 n, d_1 d_2) = 1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Lambda(n) \Phi \left(\frac{x}{p_1 p_2 n} \right) \right\} - \left\{ \frac{1}{2\pi i \varphi \left(\frac{d_1 d_2}{(d_1, d_2)} \right)} \right\} \\
&\quad \cdot \left\{ \int_{2-i\infty}^{2+i\infty} \left(1 + \frac{\omega}{(\log x)^{1.1}} \right)^{-[\log x]-1} \left(\frac{x^\omega}{\omega} \right) \sum_{\chi_{\frac{d_1 d_2}{(d_1, d_2)}} \neq \chi_0} \overline{\chi_{\frac{d_1 d_2}{(d_1, d_2)}}}(x) \frac{L'}{L}(\omega, \chi_{\frac{d_1 d_2}{(d_1, d_2)}}) \right. \\
&\quad \cdot \left. \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \chi_{\frac{d_1 d_2}{(d_1, d_2)}}(p_1 p_2) \cdot \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \left(\frac{d\omega}{(p_1 p_2)^\omega} \right) \right\}. \tag{6}
\end{aligned}$$

令

$$\begin{aligned}
M_1 &= \sum_{(d_1, x)=1} \sum_{(d_2, x)=1} \frac{\lambda_{d_1} \lambda_{d_2}}{\varphi \left(\frac{d_1 d_2}{(d_1, d_2)} \right)} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2}}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Lambda(n) \Phi \left(\frac{x}{p_1 p_2 n} \right), \\
M_2 &= \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \left| \sum_{\chi_{d^*} \neq \chi_0}^* \overline{\chi_{d^*}}(x) \int_{2-i\infty}^{2+i\infty} \left(\frac{x^\omega}{\omega} \right) \left(1 + \frac{\omega}{(\log x)^{1.1}} \right)^{-[\log x]-1} \right. \\
&\quad \cdot \left. \frac{L'}{L}(\omega, \chi_{d^*}^*) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \chi_{d^*}^*(p_1 p_2) \left((p_1 p_2)^\omega \log \frac{x}{p_1 p_2} \right)^{-1} d\omega \right|.
\end{aligned}$$

其中 d^* 是 χ_d 的 conductor, 而 $\chi_{d^*}^*$ 是等价于 χ_d 的 $\text{mod } d^*$ 的原特征. $\nu(d)$ 是 d 的素数因子的个数.

引理 5. 设 x 是偶数, 则有

$$\Omega \leq \frac{M_1 + M_2}{1 - \epsilon} + O\left(\frac{x}{(\log x)^{2.01}}\right).$$

证. 由 (5) 式和 (6) 式, 我们有

$$M \leq M_1 + |M_3| + M_4 + O\left(\frac{x}{(\log x)^{2.01}}\right), \quad (4)$$

其中

$$\begin{aligned} M_3 &= \sum_{(d_1, x)=1} \sum_{(d_2, x)=1} \frac{\lambda_{d_1} \lambda_{d_2}}{\varphi\left(\frac{d_1 d_2}{(d_1, d_2)}\right)} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2} \\ (d_1 d_2, p_1 p_2 n) > 1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}}\right) \Lambda(n) \Phi\left(\frac{x}{p_1 p_2 n}\right), \\ M_4 &= \sum_{(d_1, x)=1} \sum_{(d_2, x)=1} \left(-\frac{\lambda_{d_1} \lambda_{d_2}}{2\pi i \varphi\left(\frac{d_1 d_2}{(d_1, d_2)}\right)}\right) \int_{2-i\infty}^{2+i\infty} \left(\frac{x^\omega}{\omega}\right) \left(1 + \frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1} \\ &\quad \cdot \sum_{\substack{\chi_{\frac{d_1 d_2}{(d_1, d_2)}} \neq \chi_0}} \overline{\chi_{\frac{d_1 d_2}{(d_1, d_2)}}}(x) \frac{L'}{L}\left(\omega, \chi_{\frac{d_1 d_2}{(d_1, d_2)}}\right) \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \frac{\chi_{\frac{d_1 d_2}{(d_1, d_2)}}(p_1 p_2)}{(p_1 p_2)^\omega \log \frac{x}{p_1 p_2}} d\omega. \end{aligned}$$

首先估计 M_3 ,

$$\begin{aligned} M_3 &\ll x^\epsilon \sum_{d \leq x^{\frac{1}{2}-\epsilon}} \frac{1}{d} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2} \\ (d, p_1 p_2 n) > 1}} \Lambda(n) \ll \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{x^{1+\epsilon}}{p_1 p_2}\right) \left(\sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ p_1 | d}} \frac{1}{d}\right) \\ &\quad + \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ p_2 | d}} \frac{1}{d} \Bigg) + \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \sum_{p \leq \frac{x}{p_1 p_2}} (\log p) \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ p | d}} \frac{x^\epsilon}{d} + x^{1-\epsilon} \ll x^{1-\epsilon}. \quad (8) \end{aligned}$$

再估计 M_4 , 设 $\mu(d) \neq 0$, $d = p_1 \cdots p_k$, 则正整数 d_1 和 d_2 满足 $\frac{d_1 d_2}{(d_1, d_2)} = d$ 的充分和必要的条件是 $d_1 = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, $d_2 = p_1^{\beta_1} \cdots p_k^{\beta_k}$, 其中 $0 \leq \alpha_i \leq 1$, $0 \leq \beta_i \leq 1$, $\alpha_i + \beta_i \geq 1$ ($1 \leq i \leq k$). 故当 $d > 0$, $\mu(d) \neq 0$ 时, 则满足 $\frac{d_1 d_2}{(d_1, d_2)} = d$ 的正整数 d_1, d_2 的组数为 $3^{\nu(d)}$. 由于 $|\lambda_d| \leq 1$, 故有

$$M_4 \leq \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{3^{\nu(d)} |\mu(d)|}{\varphi(d)} \sum_{\chi_d \neq \chi_0}^* \int_{2-i\infty}^{2+i\infty} \left(\frac{x^\omega}{\omega}\right) \left(1 + \frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}$$

$$\cdot \overline{\chi_d}(x) \frac{L'}{L}(\omega, \chi_d) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \chi_d(p_1 p_2) \left(\frac{d\omega}{(p_1 p_2)^\omega \log \frac{x}{p_1 p_2}} \right) \Big|.$$

由于 $\frac{L'}{L}(\omega, \chi_d) = \frac{L'}{L}(\omega, \chi_{d^*}^*) + \sum_{p|\frac{d}{d^*}} \frac{\chi_{d^*}^*(p) \log p}{p^\omega - \chi_{d^*}^*(p)}$, 故有

$$M_4 \leq M_2 + M_5, \quad (5)$$

其中

$$M_5 = \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \left| \sum_{\chi_d \neq \chi_0}^* \overline{\chi_{d^*}^*}(x) \left(\sum_{p|\frac{d}{d^*}} \frac{\chi_{d^*}^*(p) \log p}{p^\omega - \chi_{d^*}^*(p)} \right) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \frac{\chi_{d^*}^*(p_1 p_2)}{(p_1 p_2)^\omega \log \frac{x}{p_1 p_2}} d\omega \right|.$$

又当 $\operatorname{Re} \omega = 2$ 时, 有 $\frac{\chi_{d^*}^*(p)}{p^\omega - \chi_{d^*}^*(p)} = \sum_{\lambda=1}^{\infty} \left(\frac{\chi_{d^*}^*(p)}{p^\omega} \right)^\lambda$. 又当 $\lambda \geq 1$, $\mu(d^*) \neq 0$, $(d^*, x p_1 p_2 p^\lambda) = 1$ 时, 则使用引理 4, 我们有

$$\left| \sum_{\chi_{d^*}}^* \overline{\chi_{d^*}}(x) \chi_{d^*}(p_1 p_2 p^\lambda) \right| = \left| \sum_{\chi_{d^*}}^* \chi_{d^*}(p_1 p_2 p^\lambda y) \right| \leq |(p_1 p_2 p^\lambda y - 1, d^*)| = |(x - p_1 p_2 p^\lambda, d^*)|, \quad (6)$$

其中 y 满足 $xy \equiv 1 \pmod{d^*}$ 的解。又由 (10) 式及引理 1 得到

$$\begin{aligned} M_5 &\ll \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \left| \sum_{\substack{d^*|d \\ d^* > 1}} \sum_{p|\frac{d}{d^*}} (\log p) \sum_{\lambda=1}^{\infty} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \right. \\ &\quad \left. \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Phi \left(\frac{x}{p_1 p_2 p^\lambda} \right) \right| \ll \sum_{\substack{d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \sum_{\substack{d^*|d \\ d^* > 1}} \sum_{p|\frac{d}{d^*}} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}}} \\ &\quad \cdot \sum_{1 \leq \lambda \leq \left(\log \frac{x}{p_1 p_2} \right) (\log p)^{-1}} \left(\frac{\log p}{\log \frac{x}{p_1 p_2}} \right) (x - p_1 p_2 p^\lambda, d^*) \ll \sum_{\substack{k_1, k_2 \leq x^{\frac{1}{2}-\epsilon} \\ (k_1 k_2, x)=1}} \frac{|\mu(k_1)| |\mu(k_2)| x^{\frac{\epsilon}{4}}}{\varphi(k_1) \varphi(k_2)} \\ &\quad \cdot \sum_{p, k_2} \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \sum_{1 \leq \lambda \leq \left(\log \frac{x}{p_1 p_2} \right) (\log p)^{-1}} (x - p_1 p_2 p^\lambda, k_1) \ll x^{\frac{\epsilon}{3}} \sum_{\substack{k_1 \leq x^{\frac{1}{2}-\epsilon} \\ (k_1, x)=1}} \frac{1}{k_1} \\ &\quad \cdot \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ p^\lambda \leq \frac{x}{p_1 p_2}}} (x - p_1 p_2 p^\lambda, k_1) \sum_{\substack{k_2 \leq x^{\frac{1}{2}-\epsilon} \\ k_2 \equiv 0 \pmod{p}}} \frac{1}{k_2} \ll x^{\frac{\epsilon}{2}} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ p^\lambda \leq \frac{x}{p_1 p_2}}} \\ &\quad \cdot \frac{1}{p} \sum_{d|(x - p_1 p_2 p^\lambda)} d \sum_{\substack{k_1 \leq x^{\frac{1}{2}-\epsilon} \\ d|k_1}} \frac{1}{k_1} \ll x^{1-\epsilon}. \end{aligned} \quad (11)$$

由 (7) 式, (8) 式, (9) 式及 (11) 式, 本引理得证。 $\square$

引理 6. 我们有

$$M_2 \ll \frac{x}{(\log x)^{2.01}}.$$

证. 令

$$\Phi(y, \chi) = \int_{2-i\infty}^{2+i\infty} \left(\frac{y^\omega}{\omega}\right) \left(1 + \frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1} \frac{L'}{L}(\omega, \chi) d\omega = \int_{1+\frac{1}{\log x}-i\infty}^{1+\frac{1}{\log x}+i\infty} \left(\frac{y^\omega}{\omega}\right) \left(1 + \frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]}$$

则有

$$\begin{aligned} M_2 \leq & \sum_{\substack{1 < l \leq x^{\frac{1}{2}-\epsilon} \\ (l, x)=1}} \left\{ \sum_{\substack{1 < d \leq x^{\frac{1}{2}-\epsilon} \\ l|d, (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \right\} \left| \sum_{\chi_l}^* \bar{\chi}_l(x) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \right. \\ & \cdot \Phi\left(\frac{x}{p_1 p_2}, \chi_l\right) \chi_l(p_1 p_2) \Big| \leq \sum_{\substack{1 < d \leq x^{\frac{1}{2}-\epsilon} \\ (d, x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \\ & \cdot \left\{ \sum_{\substack{1 < l \leq x^{\frac{1}{2}-\epsilon} \\ (l, x d)=1}} \frac{|\mu(l)| 3^{\nu(l)}}{\varphi(l)} \left| \sum_{\chi_l}^* \bar{\chi}_l(x) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, d)=1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Phi\left(\frac{x}{p_1 p_2}, \chi_l\right) \chi_l(p_1 p_2) \right| \right\}. \end{aligned}$$

令 $\tau(l) = \sum_{d|l} 1$, 则有

$$\sum_{1 < d \leq x^{\frac{1}{2}-\epsilon}} \frac{3^{\nu(d)} |\mu(d)|}{\varphi(d)} \ll (\log x) \sum_{d \leq x^{\frac{1}{2}-\epsilon}} \frac{(\tau(d))^2}{d} \ll (\log x)^5.$$

故有

$$M_2 \ll (\log x)^6 \max_{1 < m \leq x^{\frac{1}{2}}} N_m. \quad (7)$$

其中

$$N_m = \sum_{\substack{1 < l \leq x^{\frac{1}{2}-\epsilon} \\ (l, x)=1}} \frac{|\mu(l)| 3^{\nu(l)}}{l} \left| \sum_{\chi_l}^* \bar{\chi}_l(x) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1 p_2, m)=1}} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Phi\left(\frac{x}{p_1 p_2}, \chi_l\right) \chi_l(p_1 p_2) \right|.$$

我们用 $\sum_{(k, m)}$ 来表示一个和式, 其中的 p_1 和 p_2 经过且只经过 $x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq$

$\left(\frac{x}{p_1}\right)^{\frac{1}{2}}, x^{\frac{13}{30}} 2^k < p_1 p_2 \leq x^{\frac{13}{30}} 2^{k+1}, (p_1 p_2, m) = 1$. 令 I_1 是一个正整数, 满足 $2^{I_1-1} (\log x)^{100} < x^{\frac{1}{2}-\epsilon} < 2^{I_1} (\log x)^{100}$, $I_2 = \left\lceil \frac{7 \log x}{30 \log 2} \right\rceil$, 则有

$$N_m \leq \sum_{l=0}^{I_1} \sum_{k=0}^{I_2} N_m^{(l, k)}. \quad (8)$$

其中

$$N_m^{(0,k)} = \sum_{\substack{1 < d \leq (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \left| \sum_{\chi_d}^* \overline{\chi_d}(x) \sum_{(k,m)} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Phi \left(\frac{x}{p_1 p_2}, \chi_d \right) \chi_d(p_1 p_2) \right|.$$

而当 $l \geq 1$ 时,

$$N_m^{(l,k)} = \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \left| \sum_{\chi_d}^* \overline{\chi_d}(x) \sum_{(k,m)} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \Phi \left(\frac{x}{p_1 p_2}, \chi_d \right) \chi_d(p_1 p_2) \right|.$$

令 $S(H, \omega, \chi_d) = \sum_{n=1}^H \frac{\mu(n)\chi_d(n)}{n^\omega}$, 其中 $H \ll x$ 。我们知道当 $\operatorname{Re} \omega \geq 1$ 时, 有

$$S(H, \omega, \chi_d) \ll \log x; \quad L(\omega, \chi_d) = \sum_{n=1}^H \frac{\chi_d(n)}{n^\omega} + O\left(\frac{|\omega|d^{\frac{1}{2}} \log d}{H}\right).$$

故得到当 $\operatorname{Re} \omega \geq 1$ 时, 有

$$1 - L(\omega, \chi_d)S(H, \omega, \chi_d) = \sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^\omega} + O\left(\frac{|\omega|d^{\frac{1}{2}}(\log x)^2}{H}\right),$$

其中 $C_H(1) = 0$, 当 $n > H^2$ 时, $C_H(n) = 0$; 而当 $n > 1$ 时, $C_H(n) = -\sum_d \mu(d)$, 其中 d 经过 n 的因子, 它使得 $1 \leq d \leq H$ 及 $\frac{n}{d} \leq H$; 当 $1 \leq n \leq H$ 时, 有 $C_H(n) = 0$; 而当 $n > H$ 时, $C_H(n) \leq \tau(n)$ 。故 $H \ll x$ 时, 由 Schwarz 不等式得到

$$\left| \sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^\omega} \right|^2 \ll (\log x) \sum_{l=0}^{3I_1} \left| \sum_{n=2^l H+1}^{2^{l+1}H} \frac{C_H(n)\chi_d(n)}{n^\omega} \right|^2.$$

令 $\alpha = 1 + \frac{1}{\log x}$, 由上式、 $\sum_{n \leq x} \tau^2(n) \ll x(\log x)^3$ 及 (3) 式我们得到: 当 $Q \ll x$ 时, 有

$$\sum_{D < d \leq Q} \frac{1}{\varphi(d)} \sum_{\chi_d}^* \left| \sum_{n=2^l H+1}^{2^{l+1}H} \frac{C_H(n)\chi_d(n)}{n^{\alpha+i\nu}} \right|^2 \ll \left(Q + \frac{2^l H}{D} \right) \sum_{n=2^l H+1}^{2^{l+1}H} \frac{(\tau(n))^2}{n^2} \ll \left(\frac{Q}{2^l H} + \frac{1}{D} \right) (\log x)^3,$$

及

$$\begin{aligned} \sum_{D < d \leq Q} \frac{1}{\varphi(d)} \sum_{\chi_d}^* |1 - L(\alpha + i\nu, \chi_d)S(H, \alpha + i\nu, \chi_d)|^2 &\ll \sum_{D < d \leq Q} \frac{1}{\varphi(d)} \sum_{\chi_d}^* \left| \sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^{\alpha+i\nu}} \right|^2 + \frac{|\alpha + i\nu|^2 Q^2 (\log x)^3}{H^2} \\ &\ll \left(\frac{Q}{H} + \frac{1}{D} + \frac{|\alpha + i\nu|^2 Q^2}{H^2} \right) (\log x)^5. \end{aligned} \quad (9)$$

令 $\beta = \frac{1}{2} + \frac{1}{\log x}$, 由于 $\{S(H, \beta + i\nu, \chi_d)\}^2 = \sum_{n=1}^{H^2} \frac{j(n)\chi_d(n)}{n^{\beta+i\nu}}$, 其中 $|j(n)| \leq \tau(n)$, 故由 (3) 式可知, 当 $l \geq 1$, $H \ll x$ 时, 有

$$\sum_{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100}} \frac{1}{\varphi(d)} \sum_{\chi_d}^* |S(H, \beta + i\nu, \chi_d)|^4 \ll \left(2^l(\log x)^{100} + \frac{H^2}{2^l(\log x)^{100}} \right) \sum_{n=1}^{H^2} \frac{(\tau(n))^2}{n} \ll 2^l(\log x)^{100} \quad (10)$$

由于 $L'(\omega, \chi_d) = \frac{1}{2\pi i} \int_r \frac{L(\xi, \chi_d)}{(\xi - \omega)^2} d\xi$, 其中 r 是以 ω 为中心, $(\log x)^{-1}$ 为半径的圆, 故有

$$|L'(\omega, \chi_d)| \ll (\log x)^2 \int_r |L(\xi, \chi_d)| d\xi.$$

利用 Holder 不等式, 得到

$$|L'(\omega, \chi_d)|^4 \ll (\log x)^5 \int_r |L(\xi, \chi_d)|^4 |d\xi|.$$

又由引理 3, 我们有

$$\sum_{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100}} \left(\frac{1}{\varphi(d)} \right) \sum_{\chi_d}^* |L'(\beta + i\nu, \chi_d)|^4 \ll 2^l(\log x)^{109} (|\beta + i\nu|)^2.$$

当 $\operatorname{Re} \omega \geq \alpha = 1 + \frac{1}{\log x}$ 时, 我们得到

$$\frac{L'}{L}(\omega, \chi_d) = \left\{ \frac{L'}{L}(\omega, \chi_d) \right\} \{1 - L(\omega, \chi_d)S(H, \omega, \chi_d)\} + L'(\omega, \chi_d)S(H, \omega, \chi_d). \quad (11)$$

令

$$A(l, k, \omega, m, H) = \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100} \\ (d, x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \sum_{\chi_d}^* \left| \sum_{(k, m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^\omega \log \frac{x}{p_1 p_2}} \right| |1 - L(\omega, \chi_d)S(H, \omega, \chi_d)|,$$

$$B(l, k, \omega, m, H) = \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100} \\ (d, x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \sum_{\chi_d}^* \left| \sum_{(k, m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^\omega \log \frac{x}{p_1 p_2}} \right| |L'(\omega, \chi_d)S(H, \omega, \chi_d)|.$$

若 $l \geq 1$ 时, 由 (16) 式我们有

$$N_m^{(l, k)} \ll x(\log x)^2 \int_0^\infty \frac{A(l, k, \alpha + i\nu, m, H)}{|\alpha + i\nu| \left(1 + \frac{|\alpha + i\nu|}{(\log x)^{1.1}} \right)^{[\log x]+1}} d\nu + x^{\frac{3}{2}} \int_0^\infty \frac{B(l, k, \beta + i\nu, m, H)}{|\beta + i\nu| \left(1 + \frac{|\beta + i\nu|}{(\log x)^{1.1}} \right)^{[\log x]+1}} d\nu. \quad (12)$$

显见, 当 $|\mu(d)| \neq 0$ 及 d 很大时, 有

$$3^{\nu(d)} \leq e^{\frac{3 \log d}{\log \log d}}. \quad (13)$$

现在我们首先对 $l \geq 1$ 时, $2^k x^{\frac{13}{30}} > x^{\frac{1}{2}-\epsilon}$ 及 $x^{\frac{1}{2}-\epsilon} \geq 2^k x^{\frac{13}{30}} > 2^l (\log x)^{100}$ 这二种情形的 $N_m^{(l,k)}$ 进行估计, 此时我们取 $H = 2^l (\log x)^{200} I_{l,x}$, 其中 $I_{l,x} = e^{\frac{6 \log \{2^l (\log x)^{100}\}}{\log \log \{2^l (\log x)^{100}\}}}$ 。则根据 (14)–(18) 式, 我们有

$$\begin{aligned}
N_m^{(l,k)} &\ll x(\log x)^4 \int_0^\infty \left[\left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* \left| \sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\alpha+i\nu} \log \frac{x}{p_1 p_2}} \right|^2 \right\} \right. \\
&\quad \cdot \left. \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* |1 - L(\alpha + i\nu, \chi_d) S(H, \alpha + i\nu, \chi_d)|^2 I_{l,x} \right\}^{\frac{1}{2}} \right] \left(\frac{d\nu}{1 + \nu^{2.1}} \right) \\
&\quad + x^{\frac{3}{2}} (\log x)^4 \int_0^\infty \left\{ (I_{l,x}) \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* \left| \sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\beta+i\nu} \log \frac{x}{p_1 p_2}} \right|^2 \right\}^{\frac{1}{2}} \\
&\quad \cdot \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* |S(H, \beta + i\nu, \chi_d)|^4 \right\}^{\frac{1}{4}} \left(\frac{d\nu}{1 + \nu^4} \right) \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \right\} \\
&\ll x(\log x)^8 \int_0^\infty \left\{ \left(2^l (\log x)^{100} + \frac{2^k x^{\frac{13}{30}}}{2^l (\log x)^{100}} \right) \left(\sum_{\substack{2^k x^{\frac{13}{30}} < n \leq 2^{k+1} x^{\frac{13}{30}}} \frac{1}{n^2}} \right) \left(\frac{2^l (\log x)^{100}}{H} \right) \right. \\
&\quad \left. + \frac{1}{2^l (\log x)^{100}} + \frac{(1 + \nu^2) 2^{2l} (\log x)^{200}}{H^2} \right\}^{\frac{1}{2}} (I_{l,x}) \left(\frac{d\nu}{1 + \nu^{2.1}} \right) + x^{\frac{3}{2}} (\log x)^8 \int_0^\infty \left\{ (2^l (\log x)^{100} \right. \\
&\quad \left. + \frac{2^k x^{\frac{13}{30}}}{2^l (\log x)^{100}} \right\}^{\frac{1}{2}} (I_{l,x}) \left\{ 2^{2l} (\log x)^{213} + H^2 (\log x)^{13} \right\}^{\frac{1}{4}} (1 + \nu^2)^{\frac{1}{4}} \left(\frac{d\nu}{1 + \nu^4} \right) \ll \frac{x}{(\log x)^{20}}.
\end{aligned} \tag{19}$$

现在我们对 $2^k x^{\frac{13}{30}} \leq 2^l (\log x)^{100} \leq 2x^{\frac{1}{2}-\epsilon}$ 时的 $N_m^{(l,k)}$ 进行估计, 此时我们取

$$H = \max \left(2^{2l-k} x^{-\frac{13}{30}} (\log x)^{400} I_{l,x}, x^{\frac{1}{2}-\epsilon} \right),$$

则有

$$\begin{aligned}
N_m^{(l,k)} &\ll x(\log x)^8 \int_0^\infty \left\{ \left(2^l(\log x)^{100} + \frac{2^k x^{\frac{13}{30}}}{2^l(\log x)^{100}} \right) \left(\sum_{2^k x^{\frac{13}{30}} < n \leq 2^{k+1} x^{\frac{13}{30}}} \frac{1}{n^2} \right) \right. \\
&\quad \cdot \left(\frac{2^l(\log x)^{100}}{H} + \frac{1}{2^l(\log x)^{100}} + \frac{(1+\nu^2)2^{2l}(\log x)^{200}}{H^2} \right) (I_{l,x}) \left. \right\}^{\frac{1}{2}} \left(\frac{d\nu}{1+\nu^{2.1}} \right) \\
&\quad + x^{\frac{1}{2}}(\log x)^4 \int_0^\infty \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* |S(H, \beta + i\nu, \chi_d)|^2 \right\}^{\frac{1}{2}} (I_{l,x})^{\frac{1}{2}} \\
&\quad \cdot \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leq 2^l(\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^* \left| \left(\sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\beta + i\nu} \log \frac{x}{p_1 p_2}} \right)^2 \right|^2 \right\}^{\frac{1}{4}} \left(\frac{d\nu}{1+\nu^4} \right) \\
&\ll \frac{x}{(\log x)^{20}} + x^{\frac{1}{2}}(\log x)^{20} \left\{ 2^l(\log x)^{100} + \frac{H}{2^l(\log x)^{100}} \right\}^{\frac{1}{2}} (I_{l,x})^{\frac{3}{2}} (2^l(\log x)^{100})^{\frac{1}{4}} \\
&\quad \cdot \left(2^l(\log x)^{100} + \frac{2^{2k} x^{\frac{13}{15}}}{2^l(\log x)^{100}} \right)^{\frac{1}{4}} \int_0^\infty \frac{(1+\nu^2)^{\frac{1}{4}}}{1+\nu^4} d\nu \ll \frac{x}{(\log x)^{20}}. \tag{20}
\end{aligned}$$

现在来估计 $N_m^{(0,k)}$, 其中 $0 \leq k \leq I_2$ 。当 χ_d 是原特征及 $\operatorname{Re} S \geq 1 - \frac{c}{d^{\frac{1}{300}}}$ 时, $L(S, \chi_d) \neq 0$ 。其中 c 是一个常数, 故有

$$\begin{aligned}
N_m^{(0,k)} &\ll \sum_{1 < d \leq (\log x)^{100}} \frac{3^{\nu(d)} |\mu(d)|}{d} \sum_{\chi_d}^* \left| \int_{1 - \frac{1}{(\log x)^{1/2}} - i\infty}^{1 - \frac{1}{(\log x)^{1/2}} + i\infty} \sum_{(k,m)} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \right. \\
&\quad \cdot \chi_d(p_1 p_2) \left(\frac{x}{p_1 p_2} \right)^\omega \left(1 + \frac{\omega}{(\log x)^{1.1}} \right)^{-[\log x] - 1} \frac{L'}{L}(\omega, \chi_d) d\omega \left. \right| \\
&\ll (\log x)^{200} \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{x}{p_1 p_2} \right)^{1 - \frac{1}{(\log x)^{1/2}}} \ll \frac{x}{(\log x)^{20}}. \tag{21}
\end{aligned}$$

由 (12), (13) 式及 (19)–(21) 式, 本引理得证。 $\square$

引理 7. 对于大偶数 x , 我们有

$$M_1 \leq \left\{ \frac{(8 + 24\epsilon)x C_x}{\log x} \right\} \left\{ \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \frac{1}{p_1 p_2 \log \frac{x}{p_1 p_2}} \right\},$$

其中 $C_x = \prod_{\substack{p|x \\ p>2}} \frac{p-1}{p-2} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right)$ 。

证. 令 $S = \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} \frac{\mu^2(k)}{f(k)}$, 则有

$$\lambda_d g(d) = \left(\frac{1}{S}\right) \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k,xd)=1}} \frac{\mu(kd)\mu(k)}{f(kd)}.$$

当 $(m, x) = 1$ 时, 我们有

$$\sum_{\substack{d \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d,x)=1, m|d}} \lambda_d g(d) = \left(\frac{1}{S}\right) \left(\sum_{\substack{d \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d,x)=1, m|d}} \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k,xd)=1}} \frac{\mu(kd)\mu(k)}{f(kd)} \right) = \left(\frac{1}{S}\right) \sum_{\substack{1 \leq r \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (r,x)=1}} \frac{\mu(r)}{f(r)} \sum_{m|d|r} \mu\left(\frac{r}{d}\right) = \frac{\mu(r)}{S}.$$

由于 $\frac{1}{\varphi\left(\frac{d_1 d_2}{(d_1, d_2)}\right)} = g(d_1)g(d_2) \sum_{d|(d_1, d_2)} f(d)$. 故有

$$\begin{aligned} \sum_{\substack{d_1 \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d_1 d_2, x)=1}} \sum_{\substack{d_2 \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d_1 d_2, x)=1}} \frac{\lambda_{d_1} \lambda_{d_2}}{\varphi\left(\frac{d_1 d_2}{(d_1, d_2)}\right)} &= \sum_{d_1 \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}} \sum_{d_2 \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}} \lambda_{d_1} \lambda_{d_2} g(d_1) g(d_2) \sum_{k|(d_1, d_2)} f(k) \\ &= \sum_{\substack{k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} f(k) \left(\sum_{\substack{d \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ k|d, (d,x)=1}} \lambda_d g(d) \right)^2 = \frac{1}{S}. \end{aligned} \quad (22)$$

令 $V_k(x) = \sum_{\substack{1 \leq n \leq x \\ (n,k)=1}} \frac{\mu^2(n)}{\varphi(n)}$, 则有

$$\begin{aligned} \log x &\leq \sum_{n=1}^x \frac{1}{n} \leq \sum_{1 \leq n \leq x} \frac{\mu^2(n)}{n} \prod_{p|n} \left(\sum_{l=0}^{\infty} \frac{1}{p^l} \right) = \sum_{1 \leq n \leq x} \frac{\mu^2(n)}{n} \prod_{p|n} \left(1 - \frac{1}{p} \right)^{-1} = V_1(x) = \sum_{d|k} \sum_{\substack{1 \leq n \leq x \\ (n,k)=d}} \frac{\mu^2(n)}{\varphi(n)} \\ &= \sum_{d|k} \frac{\mu^2(d)}{\varphi(d)} \sum_{\substack{1 \leq m \leq x/d \\ (m,k)=1}} \frac{\mu^2(m)}{\varphi(m)} \leq \sum_{d|k} \frac{\mu^2(d)}{\varphi(d)} V_k(x) = \frac{k V_k(x)}{\varphi(k)}, \end{aligned}$$

故有 $V_k(x) \geq \frac{\varphi(k) \log x}{k}$. 令 $\phi(1) = 1$, 而当 $q > 2$ 时, 令 $\phi(q) = \prod_{p|q} (p-2)$, 则有

$$\begin{aligned} S &= \sum_{\substack{1 \leq k \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} \frac{\mu^2(k)}{\varphi(k)} \prod_{p|k} \left(1 + \frac{1}{p-2} \right) = \sum_{\substack{q \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (q,x)=1}} \frac{\mu^2(q)}{\phi(q)\varphi(q)} \sum_{\substack{1 \leq r \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/q \\ (r,qx)=1}} \frac{\mu^2(r)}{\varphi(r)} \\ &\geq \sum_{\substack{q \leq (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (q,x)=1}} \frac{\mu^2(q)}{\phi(q)\varphi(q)} \left\{ \frac{\varphi(qx)}{qx} \log \frac{x^{\frac{1}{4}-\frac{\epsilon}{2}}}{q} \right\} = \left(\frac{\varphi(x)}{x} \right) \left(\log x^{\frac{1}{4}-\frac{\epsilon}{2}} \right) \prod_{p|x} \left(1 + \frac{1}{p(p-2)} \right) + O(1) = \frac{\left(\frac{1}{8} - \frac{\epsilon}{4} \right)}{\prod_{p|x} \left(1 + \frac{1}{p(p-2)} \right)} + O(1) \end{aligned}$$

由 (22) 式及上式, 当 x 很大时, 有

$$M_1 \leq (8 + 24\epsilon)C_x(\log x)^{-1} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}} \\ n \leq \frac{x}{p_1 p_2}}} \left(\frac{\Lambda(n)}{\log \frac{x}{p_1 p_2}} \right) \Phi \left(\frac{x}{p_1 p_2 n} \right).$$

由引理 1, 本引理得证. □

引理 8. 设 x 是大偶数, 则有

$$\Omega \leq \frac{3.9404xC_x}{(\log x)^2}.$$

证. 当 x 很大时, 由引理 5 到引理 7, 我们有

$$\Omega \leq \left\{ \frac{8(1+5\epsilon)xC_x}{\log x} \right\} \left\{ \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \frac{1}{p_1 p_2 \log \frac{x}{p_1 p_2}} \right\}, \quad (14)$$

又有

$$\begin{aligned} & \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}} < p_2 \leq (\frac{x}{p_1})^{\frac{1}{2}}} \frac{1}{p_1 p_2 \log \frac{x}{p_1 p_2}} \leq (1+\epsilon) \sum_{x^{\frac{1}{10}} < p_1 \leq x^{\frac{1}{3}}} \int_{x^{\frac{1}{3}}}^{(\frac{x}{p_1})^{\frac{1}{2}}} \frac{dt}{p_1 t (\log t) \log \frac{x}{p_1 t}} \\ & \leq (1+2\epsilon) \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S \log S} \int_{S^{\frac{1}{3}}}^{(\frac{x}{S})^{\frac{1}{2}}} \frac{dt}{t (\log t) \left(\log \frac{x}{St} \right)} = (1+2\epsilon) \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha} \int_{\frac{1}{3}}^{\frac{1-\alpha}{2}} \frac{d\beta}{\beta(1-\alpha-\beta) \log x}, \end{aligned}$$

及

$$\begin{aligned} & \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha} \int_{\frac{1}{3}}^{\frac{1-\alpha}{2}} \left(\frac{1}{1-\alpha} \right) \left(\frac{1}{\beta} + \frac{1}{1-\alpha-\beta} \right) d\beta = \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{\log \frac{1-\alpha}{2} - \log \frac{1}{3} - \log \frac{1-\alpha}{2} + \log \left(\frac{2}{3} - \alpha \right)}{\alpha(1-\alpha)} d\alpha \\ & = \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{\log(2-3\alpha)}{\alpha(1-\alpha)} d\alpha = \sum_{i=0}^6 \int_{\frac{1}{10} + \frac{i}{30}}^{\frac{1}{10} + \frac{i+1}{30}} \frac{\log \left(1.6 - \frac{i}{10} \right)}{\alpha(1-\alpha)} d\alpha + \sum_{i=0}^6 \int_{\frac{1}{10} + \frac{i}{30}}^{\frac{1}{10} + \frac{i+1}{30}} \frac{\log \frac{2-3\alpha}{1.6-0.i}}{\alpha(1-\alpha)} d\alpha \\ & \leq \sum_{i=0}^6 \left\{ \log \left(1.6 - \frac{i}{10} \right) \right\} \left\{ \log \frac{\frac{9}{10} - \frac{i}{30}}{\frac{1}{10} + \frac{i}{30}} - \log \frac{\frac{9}{10} - \frac{1}{30} - \frac{i}{30}}{\frac{1}{10} + \frac{i+1}{30}} \right\} \\ & + \sum_{i=0}^6 \int_{\frac{1}{10} + \frac{i}{30}}^{\frac{1}{10} + \frac{i+1}{30}} \frac{(0.4 + 0.i - 3\alpha)}{(1.6 - 0.i)\alpha(1-\alpha)} d\alpha \leq \sum_{i=0}^6 \left\{ \log(1.6 - 0.i) + \frac{4+i}{16-i} \right\} \left\{ \log \frac{27-i}{3+i} \right. \\ & \left. - \log \frac{26-i}{4+i} \right\} - 3 \sum_{i=0}^6 \int_{\frac{1}{10} + \frac{i}{30}}^{\frac{1}{10} + \frac{i+1}{30}} \frac{d\alpha}{(1.6 - 0.i)(1-\alpha)} = \sum_{i=0}^6 \left\{ \log(1.6 - 0.i) + \frac{4+i}{16-i} \right\} \end{aligned}$$

$$\begin{aligned}
& \cdot \left\{ \log \frac{108 + 23i - i^2}{78 + 23i - i^2} \right\} - 3 \sum_{i=0}^6 \left(\frac{1}{1.6 - 0.i} \right) \left(\log \frac{27 - i}{26 - i} \right) \\
& \leq (0.47+0.25)(0.32542)+(0.40547+0.33334)(0.26236)+(0.33647+0.42858)(0.22315)+(0.26236+0.53847) \\
& + (0.18232+0.66667)(0.17799)+(0.09531+0.81819)(0.16431)+0.15415-3 \left(\frac{0.03774}{1.6} + \frac{0.03922}{1.5} + \frac{0.04082}{1.4} \right. \\
& \left. + \frac{0.04445}{1.2} + \frac{0.04652}{1.1} + 0.04879 \right) \leq 0.234303+0.193837+0.17073+0.15754+0.151115+0.1501+0.15415 \\
& -3(0.023587+0.026146+0.029157+0.032738+0.037041+0.04229+0.04879) \leq 1.21178-0.71924 = 0.49254 \\
& \quad (15)
\end{aligned}$$

由 (23) 和 (24) 式, 引理 8 得证。 $\square$

设 x 是一大偶数, 令 $P_x(x, x^{\frac{1}{10}})$ 表示满足下面条件的素数 p 的个数: $p \leq x$, $p \not\equiv x \pmod{p_i}$ ($1 \leq i \leq j$), 其中 $3 = p_1 < p_2 < \cdots < p_j \leq x^{\frac{1}{10}}$ 。对于一个素数 p' , 则令 $P_x(x, p', x^{\frac{1}{10}})$ 表示满足下面条件的素数 p 的个数: $p \leq x$, $p \equiv x \pmod{p'}$, $p \not\equiv x \pmod{p_i}$ ($1 \leq i \leq j$)。其中 $3 = p_1 < p_2 < \cdots < p_j \leq x^{\frac{1}{10}}$ 。

引理 9. 设 x 是大偶数, 则有

$$P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2} \right) \sum_{x^{\frac{1}{10}} < p \leq x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) \geq \frac{2.6408x C_x}{(\log x)^2},$$

$$\text{其中 } C_x = \prod_{\substack{p|x \\ p>2}} \frac{p-1}{p-2} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right).$$

证. 在文献 [11] 中取 $r(p) = \frac{p}{p-1}$, $K = x$, $Z = x^{\frac{1}{10}}$, 则显见文献 [11] 中的条件 (A_1) 和 (A_2) 都满足, 由文献 [11] 中的 (2.11) 式, 我们有

$$\begin{aligned}
\Gamma_x(x^{\frac{1}{10}}) &= \frac{x}{\varphi(x)} \prod_{p|x} \frac{1 - \frac{1}{p-1}}{1 - \frac{1}{p}} \cdot \frac{e^{-r}}{\log x^{\frac{1}{10}}} \left\{ 1 + O\left(\frac{1}{\log x} \right) \right\} \\
&= \frac{x}{\varphi(x)} \prod_{\substack{p|x \\ p>2}} \frac{(p-1)^2}{p(p-2)} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right) \frac{e^{-r}}{\log x^{\frac{1}{10}}} \left\{ 1 + O\left(\frac{1}{\log x} \right) \right\} = \frac{20e^{-r}C_x}{\log x} \left\{ 1 + O\left(\frac{1}{\log x} \right) \right\}.
\end{aligned} \quad (25)$$

其中 r 是 Euler 常数。又当 $0 < u \leq 2$ 时, 令 $F(u) = \frac{2e^r}{u}$, $f(u) = 0$ 。而当 $u \geq 2$ 时, 令 $(uF(u))' = f(u-1)$, $(uf(u))' = F(u-1)$, 当 $2 < u \leq 3$ 时, 有 $uF(u) = 2F(2)$, $F(u) = \frac{2e^r}{u}$ 。又当 $2 < u \leq 4$ 时, 则有

$$uf(u) = \int_2^u F(t-1) dt = 2e^r \log(u-1), \quad f(u) = \frac{2e^r \log(u-1)}{u}.$$

当 $3 \leq u \leq 4$ 时, 我们有

$$uF(u) = 2e^r + \int_3^u f(t-1) dt = 2e^r \left(1 + \int_2^{u-1} \frac{\log(t-1)}{t} dt \right),$$

又有

$$5f(5) = 2e^r \log 3 + \int_4^5 F(u-1) du = 2e^r \left(\log 4 + \int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t} dt \right).$$

在文献 [11] 的定理 A 中, 取 $\xi^2 = x^{\frac{1}{2}-\epsilon}$, $q = 1$, $z = x^{\frac{1}{10}}$, 则由 (25) 式及文献 [11] 中的 (2.19), (4.18) 及 (3.24) 式, 我们知道当 x 很大时, 有

$$P_x(x, x^{\frac{1}{10}}) \geq \frac{2(1-\sqrt{\epsilon})e^{-r}xC_x f(5)}{(\log x)(\log x^{\frac{1}{10}})} \geq \left\{ \frac{8(1-\sqrt{\epsilon})xC_x}{(\log x)^2} \right\} \left\{ \log 4 + \int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t} dt \right\}. \quad (26)$$

又在文献 [11] 的定理 A 中取 $\xi^2 = \frac{x^{\frac{1}{2}-\epsilon}}{p}$, $q = p$, $z = x^{\frac{1}{10}}$, 则由 (25) 式及文献 [11] 中的 (2.18), (3.24) 及 (4.18), 我们有

$$\begin{aligned} \sum_{x^{\frac{1}{10}} < p \leq x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) &\leq \left\{ \frac{20(1+\sqrt{\epsilon})e^{-r}xC_x}{(\log x)^2} \right\} \left\{ \sum_{x^{\frac{1}{10}} < p \leq x^{\frac{1}{5}}} \left(\frac{2e^r}{p} \right) \left(1 + \int_2^{4-\frac{10\log p}{\log x}} \frac{\log(t-1)}{t} dt \right) (\log x^{\frac{1}{10}}) \right. \\ &\quad \left. + \sum_{x^{\frac{1}{5}} < p \leq x^{\frac{1}{3}}} \frac{2e^r \log x^{\frac{1}{10}}}{p \log \frac{x^{\frac{1}{2}}}{p}} \right\} \leq \left\{ \frac{(4+5\sqrt{\epsilon})xC_x}{\log x} \right\} \left\{ \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{5}}} \frac{dS}{S(\log S) \left(\log \frac{x^{\frac{1}{2}}}{S} \right)} \int_2^{4-\frac{10\log S}{\log x}} \frac{\log(t-1)}{t} dt + \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha} \right. \\ &\quad \left. \int_2^{4-10\alpha} \frac{\log(t-1)}{t} dt \right\}. \end{aligned}$$

令 $4-10\alpha = u-1$, $\alpha = \frac{5-u}{10}$, $\frac{d\alpha}{\alpha \left(\frac{1}{2} - \alpha \right)} = -\frac{10 du}{u(5-u)}$, 又当 $\alpha = \frac{1}{10}$ 时, 有

$u = 4$, 而当 $\alpha = \frac{1}{5}$ 时, $u = 3$, 故有

$$\int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha \left(\frac{1}{2} - \alpha \right)} \int_2^{4-10\alpha} \frac{\log(t-1)}{t} dt = \int_3^4 \frac{10 du}{u(5-u)} \int_2^{u-1} \frac{\log(t-1)}{t} dt.$$

显见, 当 $1 \leq x \leq 2$ 时, 有 $\log x \leq \frac{x-1}{2} + \frac{x-1}{1+x}$, 故有

$$\begin{aligned} &\int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t} dt - \left(\frac{1}{4} \right) \int_{\frac{1}{10}}^{\frac{1}{5}} \frac{d\alpha}{\alpha \left(\frac{1}{2} - \alpha \right)} \int_2^{4-10\alpha} \frac{\log(t-1)}{t} dt \\ &= \int_3^4 \left(\frac{1}{u} - \frac{2.5}{u(5-u)} \right) du \int_2^{u-1} \frac{\log(t-1)}{t} dt \geq \int_3^4 \left\{ \frac{2.5-u}{u(5-u)} \right\} du \int_2^{u-1} \left(\frac{t-2}{2} + \frac{t-2}{t} \right) \left(\frac{dt}{t} \right) \end{aligned}$$

$$\begin{aligned}
&= \int_3^4 \left\{ \frac{2.5-u}{2u(5-u)} \right\} \left(u - 3 + \frac{4}{u-1} - 2 \right) du = \int_3^4 \left(\frac{1}{2} - \frac{2.25}{u} - \frac{1}{4(5-u)} + \frac{0.75}{u-1} \right) du \\
&= \frac{1}{2} - 2.25 \log \frac{4}{3} - \frac{\log 2}{4} + 0.75 \log \frac{3}{2} = \frac{1}{2} + 0.75 \log \frac{9}{8} - 1.5 \log \frac{4}{3} - \frac{\log 2}{4} \\
&\geq 0.588335 - 0.6048075 = -0.0164725.
\end{aligned} \tag{16}$$

由 (26) 和 (27) 式, 我们有

$$P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2} \right) \sum_{x^{\frac{1}{10}} < p \leq x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) \geq \left(\frac{(8 - 50\sqrt{\epsilon})xC_x}{(\log x)^2} \right) \left(\log 4 - \frac{\log 8}{2} - 0.0164725 \right) \geq \frac{(8xC_x)(0.3)}{(\log x)}$$

故引理 9 得证。 $\square$

3 结果

显见, 我们有

$$P_x(1, 2) \geq P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2} \right) \sum_{x^{\frac{1}{10}} < p \leq x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) - \frac{\Omega}{2} - x^{0.91}. \tag{17}$$

由 (28) 式、引理 8 和引理 9, 即得到定理 1

$$(1, 2) \text{ 及 } P_x(1, 2) \geq \frac{0.67xC_x}{(\log x)^2}$$

的证明。

完全类似的方法可得到定理 2 的证明。

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